

MACHINE PROBLEM 1

1. CSV READER

The KNIME Analytics Platform interface displays a workflow for processing CSV data. The workflow starts with a 'CSV Reader' node, followed by a 'Column Filter', 'Normalizer', and 'Table Partitioner'. The 'Table Partitioner' outputs are connected to four machine learning nodes: 'Decision Tree Learner', 'Logistic Regression Learner', 'Random Forest Learner', and 'Scorer'. The 'Scorer' nodes are connected to 'Scorer' nodes. The 'CSV Reader' node is configured with the following settings:

- File: C:\Users\Dianna\Downloads\student_performance.csv
- File Format: Skip first lines of file: 0
- Comment line character: #
- Row delimiter: Line break
- Column delimiter: ,
- Quoted strings contain no row delimiters: ☐
- Quote character: "
- Quote escape character: \
- Autodetect format: ☒

The 'Table' view shows the following data:

#	RowID	student_id	attendance	quiz_score	assignment_score	exam_score	risk_status
1	Row0	S001	95	90	92	88	Not At Risk
2	Row1	S002	88	84	86	82	Not At Risk
3	Row2	S003	76	72	75	70	Not At Risk
4	Row3	S004	62	58	60	55	At Risk
5	Row4	S005	55	48	52	50	At Risk
6	Row5	S006	91	87	89	85	Not At Risk

2. COLUMN FILTER

The KNIME Analytics Platform interface displays a workflow for processing CSV data. The workflow starts with a 'CSV Reader' node, followed by a 'Column Filter', 'Normalizer', and 'Table Partitioner'. The 'Table Partitioner' outputs are connected to four machine learning nodes: 'Decision Tree Learner', 'Logistic Regression Learner', 'Random Forest Learner', and 'Scorer'. The 'Scorer' nodes are connected to 'Scorer' nodes. The 'Column Filter' node is configured with the following settings:

- Column filter: Manual
- Excludes: student_id
- Includes: attendance, quiz_score, assignment_score, exam_score, risk_status

The 'Table' view shows the following data:

#	RowID	attendance	quiz_score	assignment_score	exam_score	risk_status
1	Row0	95	90	92	88	Not At Risk
2	Row1	88	84	86	82	Not At Risk
3	Row2	76	72	75	70	Not At Risk

3. NORMALIZER

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Deploy on Hub

CSV Reader Column Filter Normalizer Table Partitioner Decision Tree Learner Decision Tree Predictor Scorer Logistic Regression Learner Logistic Regression Predictor Scorer Random Forest Learner Random Forest Predictor Scorer

1: Normalized table 2: Normalize Model Flow Variables

Rows: 30 Columns: 5 Sorting Filter

#	RowID	attendance	quiz_score	assignment_score	exam_score	risk_status
1	Row0	0.98	0.92	0.979	0.868	Not At Risk
2	Row1	0.837	0.8	0.854	0.755	Not At Risk
3	Row2	0.592	0.56	0.625	0.528	Not At Risk

4. TABLE PARTITIONER

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CSV Reader Column Filter Normalizer Table Partitioner Decision Tree Learner Decision Tree Predictor Scorer Logistic Regression Learner Logistic Regression Predictor Scorer Random Forest Learner Random Forest Predictor Scorer

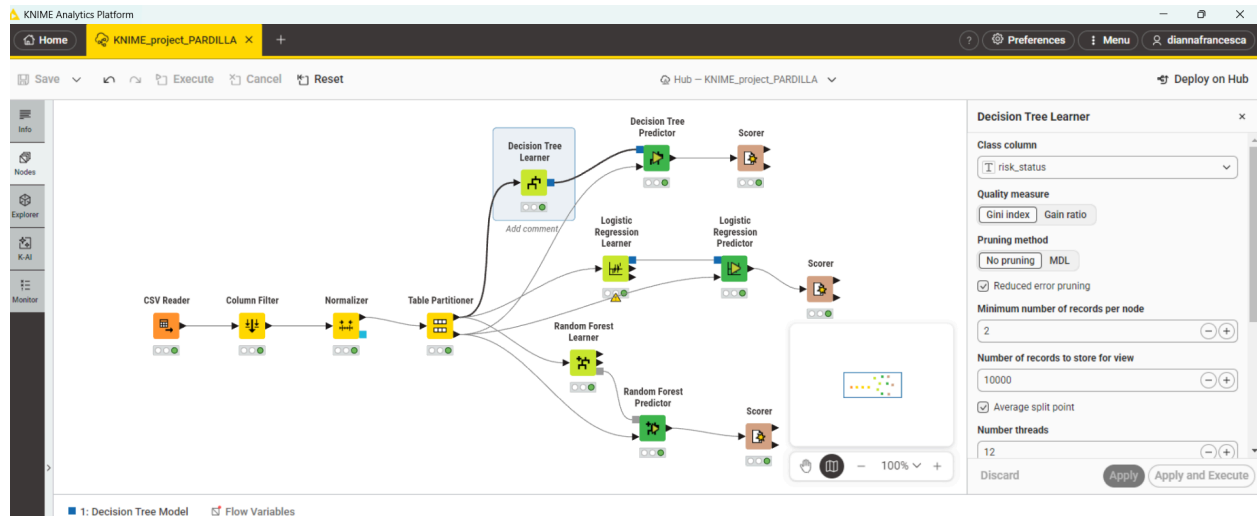
1: First partition 2: Second partition Flow Variables

Rows: 24 Columns: 5 Sorting Filter

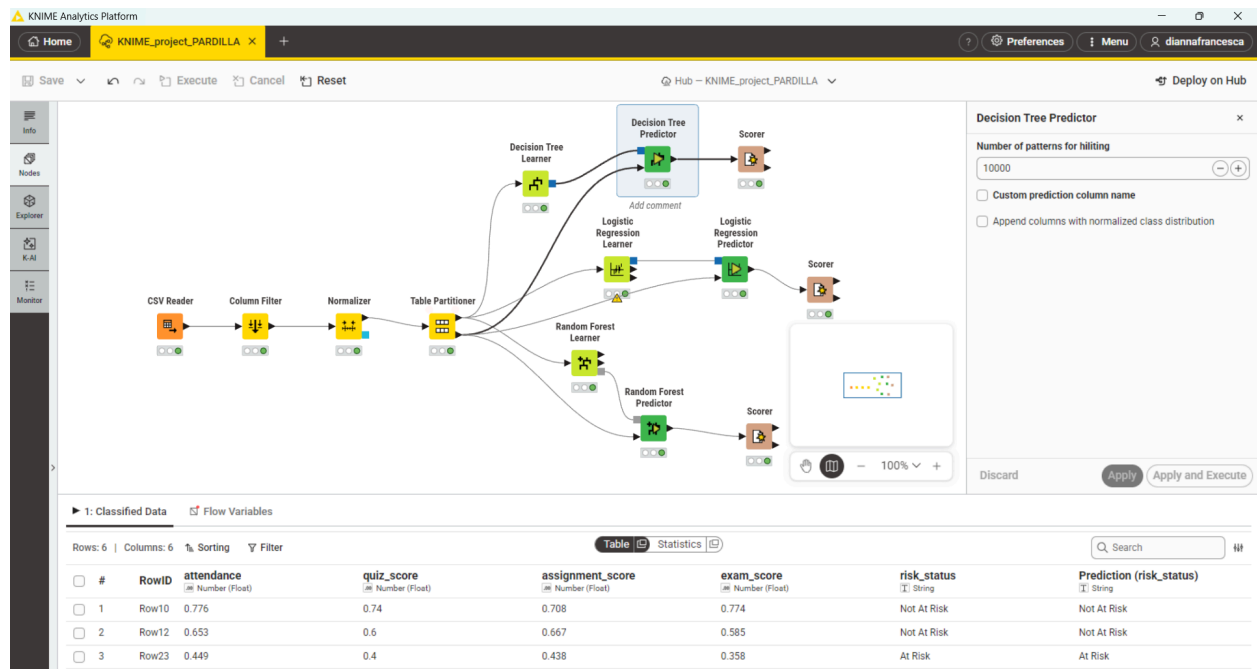
#	RowID	attendance	quiz_score	assignment_score	exam_score	risk_status
1	Row0	0.98	0.92	0.979	0.868	Not At Risk
2	Row1	0.837	0.8	0.854	0.755	Not At Risk
3	Row2	0.592	0.56	0.625	0.528	Not At Risk

5. DECISION TREE ALGORITHM

a. LEARNER



b. PREDICTOR



c. SCORER

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Scorer

First column: Prediction (risk_status)

Second column: risk_status

Sorting strategy: Insertion order

Missing values: Ignore, Fail

Discard Apply Apply and Execute

1: Confusion matrix 2: Accuracy statistics Flow Variables

#	RowID	Not At Risk	At Risk
1	Not A...	3	0
2	At Risk	0	3

6. LOGISTIC REGRESSION ALGORITHM

a. LEARNER

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Logistic Regression Learner

Target column: risk_status

Reference category: At Risk

Solver: Stochastic average gradient

Feature selection: Manual

Includes: attendance, quiz_score, assignment_s...

Discard Apply Apply and Execute

1: Model for Predictor 2: Coefficients and Statistics 3: Model and Learning Properties Flow Variables

The data at the output port is not supported by any modern viewer.

b. PREDICTOR

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CSV Reader Column Filter Normalizer Table Partitioner

Decision Tree Learner Decision Tree Predictor Scorer

Logistic Regression Learner Logistic Regression Predictor Scorer

Random Forest Learner Random Forest Predictor Scorer

Add comment

Logistic Regression Predictor

Custom prediction column name

Append columns with normalized class distribution

Discard Apply Apply and Execute

► 1: Predicted data

#	RowID	attendance	quiz_score	assignment_score	exam_score	risk_status	Prediction (risk_status)
1	Row10	0.776	0.74	0.708	0.774	Not At Risk	Not At Risk
2	Row12	0.653	0.6	0.667	0.585	Not At Risk	Not At Risk
3	Row23	0.449	0.4	0.438	0.358	At Risk	At Risk

c. SCORER

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Info Nodes Explorer K.AI Monitor

CSV Reader Column Filter Normalizer Table Partitioner

Decision Tree Learner Decision Tree Predictor Scorer

Logistic Regression Learner Logistic Regression Predictor Scorer

Random Forest Learner Random Forest Predictor Scorer

Scorer

First column

Prediction (risk_status)

Second column

risk_status

Sorting strategy

Insertion order

Reverse order

Use name prefix

Missing values

Ignore Fail

Discard Apply Apply and Execute

► 1: Confusion matrix

#	RowID	Not At Risk	At Risk
1	Not A...	3	0
2	At Risk	0	3

7. RANDOM FOREST ALGORITHM

a. LEARNER

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Info Nodes Explorer K-4J Monitor

CSV Reader Column Filter Normalizer Table Partitioner

Decision Tree Learner Decision Tree Predictor Scorer

Logistic Regression Learner Logistic Regression Predictor Scorer

Random Forest Learner Random Forest Predictor Scorer

Random Forest Learner

Attribute Selection

Target column: risk_status

Training attributes: Use column Use fingerprint

Attribute selection: Manual Pattern

Search

Excludes: No columns in this list.

Includes: attendance, quiz_score, assignment_s..., exam_score

Discard Apply Apply and Execute

► 1: Out-of-bag Predictions ► 2: Attribute Statistics ► 3: Random Forest Model ► Flow Variables

Table Statistics

Rows: 24 Columns: 10 Sorting Filter

#	RowID	attendance	quiz_score	assignment_score	exam_score	risk_status	P (risk_status=N...	P (risk_status=A...	risk_status (Out...	risk_status (Out...	model count
1	Row0	0.98	0.92	0.979	0.868	Not At Risk					
2	Row1	0.837	0.8	0.854	0.755	Not At Risk					
3	Row2	0.592	0.56	0.625	0.528	Not At Risk					

b. PREDICTOR

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CSV Reader Column Filter Normalizer Table Partitioner

Decision Tree Learner Decision Tree Predictor Scorer

Logistic Regression Learner Logistic Regression Predictor Scorer

Random Forest Learner Random Forest Predictor Scorer

Random Forest Predictor

Change prediction column name

Prediction column name: Prediction (risk_status)

Append overall prediction confidence

Append individual class probabilities

Suffix for probability columns

Use soft voting

Discard Apply Apply and Execute

► 1: Prediction output ► Flow Variables

Table Statistics

Rows: 6 Columns: 7 Sorting Filter

#	RowID	attendance	quiz_score	assignment_score	exam_score	risk_status	Prediction (risk_status)	Prediction (risk_status) (...)
1	Row10	0.776	0.74	0.708	0.774	Not At Risk	Not At Risk	1
2	Row12	0.653	0.6	0.667	0.585	Not At Risk	Not At Risk	1
3	Row23	0.449	0.4	0.438	0.358	At Risk	At Risk	1

c. SCORER

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Info Nodes Explorer KAI Monitor

CSV Reader Column Filter Normalizer Table Partitioner

Decision Tree Learner Decision Tree Predictor Scorer

Logistic Regression Learner Logistic Regression Predictor Scorer

Random Forest Learner Random Forest Predictor Scorer

Scorer

First column: Prediction (risk_status)

Second column: risk_status

Sorting strategy: Insertion order

Reverse order: ☐

Use name prefix: ☐

Missing values: Ignore Fail

Discard Apply Apply and Execute

► 1: Confusion matrix ► 2: Accuracy statistics Flow Variables

Rows: 2 Columns: 2 Sorting Filter Table Statistics Search

#	RowID	Not At Risk	At Risk
		Number (Integer)	Number (Integer)
☐	1	Not A...	3
☐	2	At Risk	0